Lab Sheet: 1

Objective:

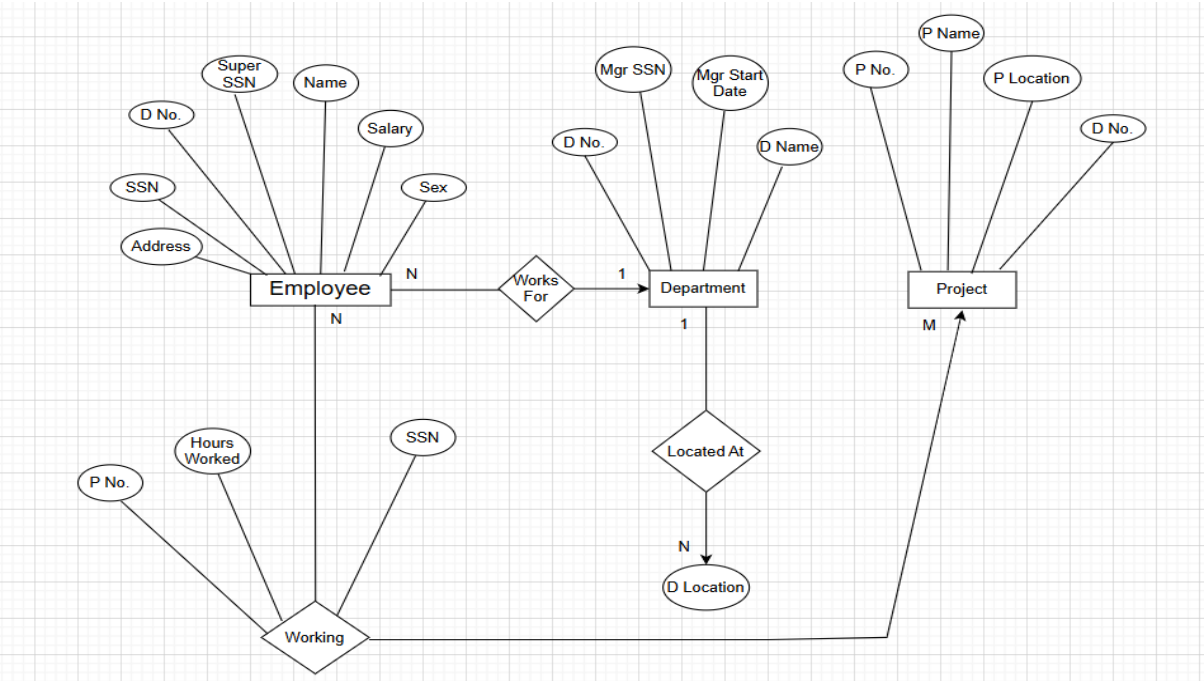
- To design ER diagrams and convert entities and relationships into relational tables using proper schema notation.

Instructions:

1. Open your preferred Database Management System (DBMS) tool.
2. Create a new database named DBMS_YourName.
3. Draw ER diagrams for the given scenarios.
4. Convert entities and relationships into relational schemas.
5. Create tables using the derived schemas.

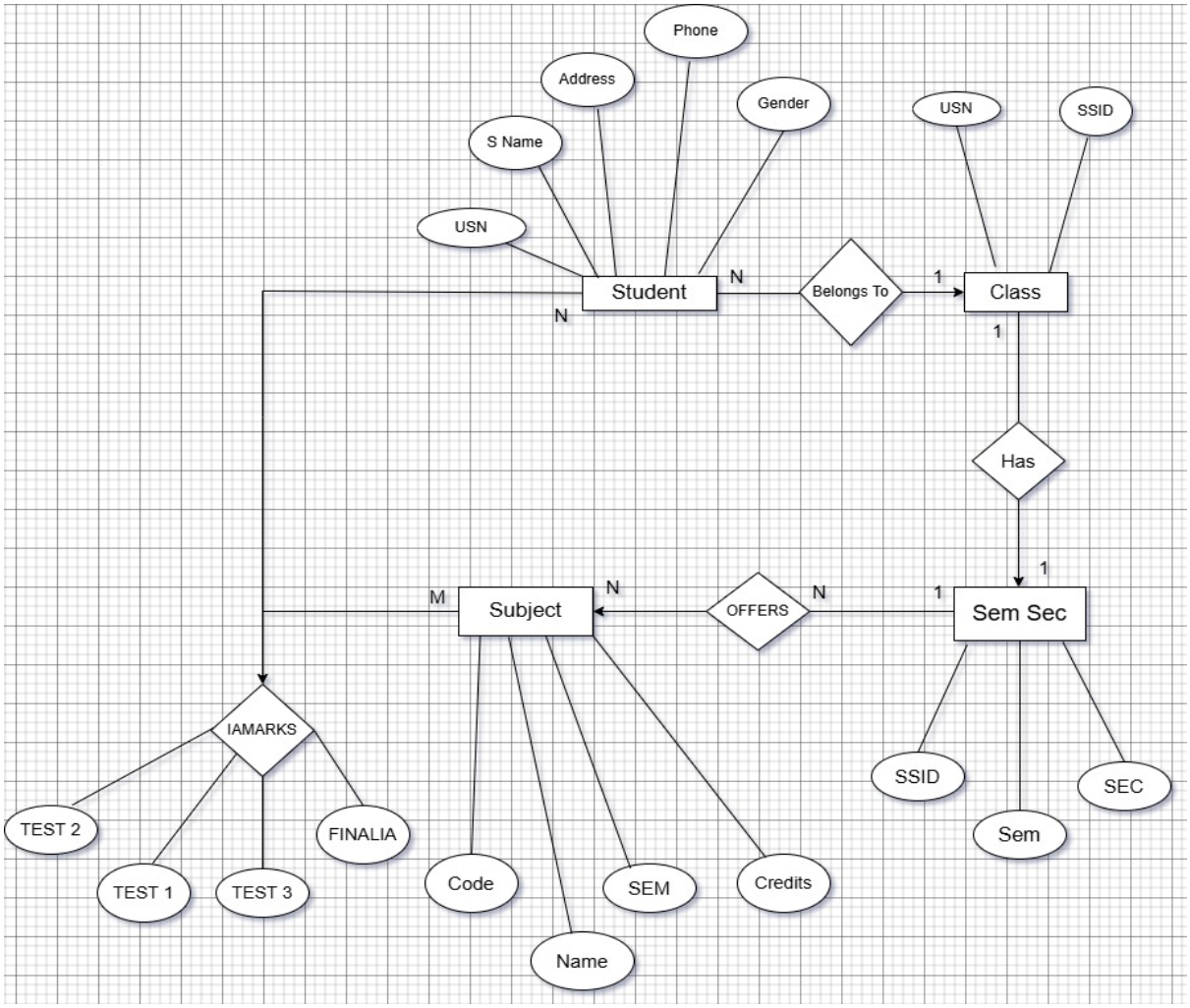
PART A – COLLEGE DATABASE

STUDENT (USN, SName, Address, Phone, Gender)
SEMSEC (SSID, Sem, Sec)
CLASS (USN, SSID)
SUBJECT (Subcode, Title, Sem, Credits)
IAMARKS (USN, Subcode, SSID, Test1, Test2, Test3, FinalIA)



PART B – COMPANY DATABASE

EMPLOYEE (SSN, Name, Address, Sex, Salary, Superssn, Dno)
DEPARTMENT (Dno, Dname, Mgrssn, Mgrstartdate)
DLOCATION (Dno, Dloc)
PROJECT (PNo, PName, PLocation, DNo)
WORKS_ON (SSN, PNo, Hours)



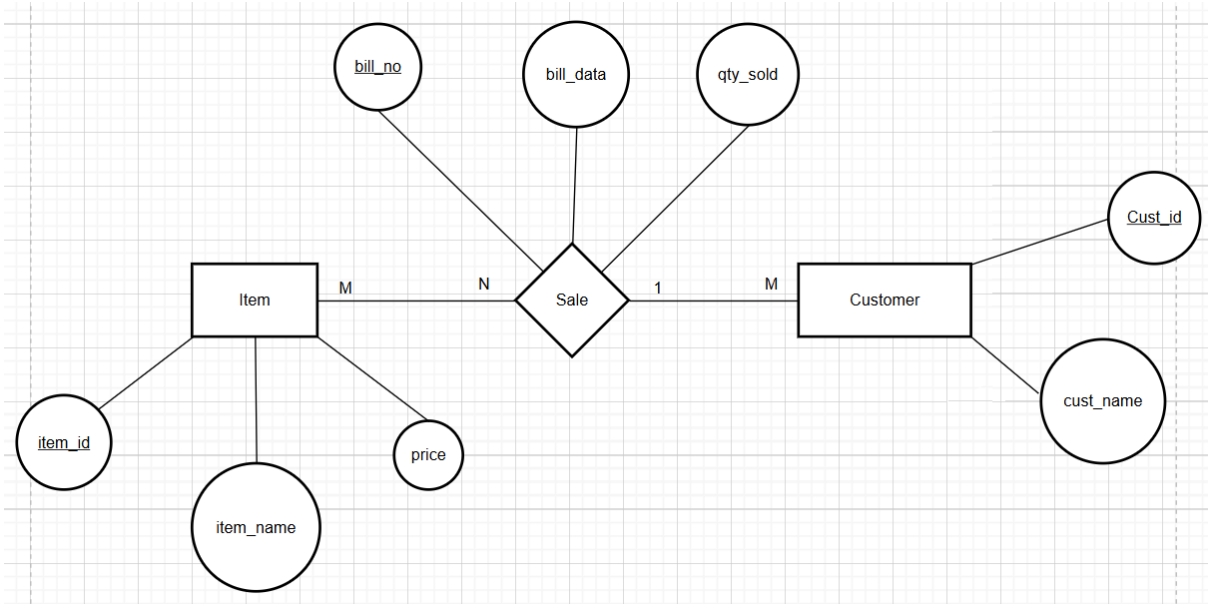
PART C – CUSTOMER SALE SCHEMA

Database Schema for a customer-sale scenario

Customer(Cust id : integer, cust_name: string)

Item(item_id: integer, item_name:string, price: integer)

Sale(bill_no:integer,bill_data:date,cust_id: integer, item_id: integer, qty_sold: integer)



DDL Questions (Data Definition Language):

1. Write SQL statements to create all the above relations with appropriate PRIMARY KEY and FOREIGN KEY constraints.

```
CREATE database DBMS_Ishaan;

CREATE TABLE STUDENT (

  USN VARCHAR(10) PRIMARY KEY,

  SName VARCHAR(50) NOT NULL,

  Address VARCHAR(100),

  Phone VARCHAR(15),

  Gender CHAR(1) CHECK (Gender IN ('M', 'F', 'O'))

);

CREATE TABLE SEMSEC (

  SSID VARCHAR(10) PRIMARY KEY,

  Sem INT CHECK (Sem BETWEEN 1 AND 8),

  Sec CHAR(1)

);

CREATE TABLE CLASS (

  USN VARCHAR(10),
```

```
  SSID VARCHAR(10),
  PRIMARY KEY (USN, SSID),
  FOREIGN KEY (USN) REFERENCES STUDENT(USN),
  FOREIGN KEY (SSID) REFERENCES SEMSEC(SSID)
);
CREATE TABLE SUBJECT (
  Subcode VARCHAR(10) PRIMARY KEY,
  Title VARCHAR(100) NOT NULL,
  Sem INT,
  Credits INT
);
CREATE TABLE IAMARKS (
  USN VARCHAR(10),
  Subcode VARCHAR(10),
  SSID VARCHAR(10),
  Test1 INT DEFAULT 0,
  Test2 INT DEFAULT 0,
  Test3 INT DEFAULT 0,
  FinalIA INT,
  PRIMARY KEY (USN, Subcode, SSID),
  FOREIGN KEY (USN) REFERENCES STUDENT(USN) ,
  FOREIGN KEY (Subcode) REFERENCES SUBJECT(Subcode) ,
  FOREIGN KEY (SSID) REFERENCES SEMSEC(SSID)
);
CREATE TABLE DEPARTMENT (
  Dno INT PRIMARY KEY,
  Dname VARCHAR(50) NOT NULL,
  Mgrssn CHAR(9),
```

```
    Mgrstartdate DATE
);
CREATE TABLE EMPLOYEE (
    SSN CHAR(9) PRIMARY KEY,
    Name VARCHAR(50) NOT NULL,
    Address VARCHAR(100),
    Sex CHAR(1) CHECK (Sex IN ('M', 'F')),
    Salary DECIMAL(10, 2),
    Superssn CHAR(9),
    Dno INT,
    FOREIGN KEY (Superssn) REFERENCES EMPLOYEE(SSN),
    FOREIGN KEY (Dno) REFERENCES DEPARTMENT(Dno)
);
ALTER TABLE DEPARTMENT
ADD CONSTRAINT FK_Mgr
FOREIGN KEY (Mgrssn) REFERENCES EMPLOYEE(SSN);
CREATE TABLE DLOCATION (
    Dno INT,
    Dloc VARCHAR(50),
    PRIMARY KEY (Dno, Dloc),
    FOREIGN KEY (Dno) REFERENCES DEPARTMENT(Dno)
);
CREATE TABLE PROJECT (
    PNo INT PRIMARY KEY,
    PName VARCHAR(50) NOT NULL,
    PLocation VARCHAR(50),
    DNo INT,
    FOREIGN KEY (DNo) REFERENCES DEPARTMENT(Dno)
```

```
);
CREATE TABLE WORKS_ON (
   SSN CHAR(9),
   PNo INT,
   Hours DECIMAL(4, 1),
   PRIMARY KEY (SSN, PNo),
   FOREIGN KEY (SSN) REFERENCES EMPLOYEE(SSN) ,
   FOREIGN KEY (PNo) REFERENCES PROJECT(PNo)
);
CREATE TABLE Customer (
   Cust_id INT PRIMARY KEY,
   cust_name VARCHAR(100) NOT NULL
);

CREATE TABLE Item (
   item_id INT PRIMARY KEY,
   item_name VARCHAR(100) NOT NULL,
   price INT
);

CREATE TABLE Sale (
   bill_no INT,
   bill_data DATE,
   cust_id INT,
   item_id INT,
   qty_sold INT,
   -- Composite Primary Key: Allows multiple items per bill
   PRIMARY KEY (bill_no, item_id),
```

```
  FOREIGN KEY (cust_id) REFERENCES Customer(Cust_id),
  FOREIGN KEY (item_id) REFERENCES Item(item_id)
);
```

2. Add a new column Email to the STUDENT relation.

```
ALTER TABLE STUDENT
ADD Email VARCHAR(100);
```

	USN	SName	Address	Phone	Gender	Email
*	NULL	NULL	NULL	NULL	NULL	NULL

3. Apply NOT NULL constraints to Salary and Price attributes

```
ALTER TABLE EMPLOYEE
MODIFY Salary DECIMAL(10, 2) NOT NULL;
```

	SSN	Name	Address	Sex	Salary	Superssn	Dno
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL

4. Rename attribute SName to StudentName in STUDENT relation

```
ALTER TABLE STUDENT
RENAME COLUMN SName TO StudentName;
```

	USN	StudentName	Address	Phone	Gender	Email
*	NULL	NULL	NULL	NULL	NULL	NULL

5. Drop the DLOCATION relation.

```
DROP TABLE DLOCATION;
```

	Tables_in_dbms_ishaan
▶	class
	customer
	department
	employee
	iamarks
	item
	project
	sale
	semsec
	student
	subject
	works_on

DML Questions (Data Manipulation Language):

1. Insert at least 10 tuples into CUSTOMER, ITEM, and SALE relations.

```
INSERT INTO Customer (Cust_id, cust_name) VALUES
(1, 'Amit Sharma'),
(2, 'Priya Singh'),
(3, 'John Doe'),
(4, 'Sita Verma'),
(5, 'Rajesh Kumar'),
(6, 'Ananya Iyer'),
(7, 'Vikram Rathore'),
(8, 'Megha Gupta'),
(9, 'Suresh Raina'),
(10, 'Kavita Reddy');

INSERT INTO Item (item_id, item_name, price) VALUES
(101, 'Laptop', 55000),
(102, 'Smartphone', 25000),
(103, 'Headphones', 2500),
(104, 'Smartwatch', 5000),
(105, 'Keyboard', 1200),
(106, 'Gaming Mouse', 1800),
(107, 'Monitor', 12000),
(108, 'USB Drive 64GB', 800),
(109, 'External Hard Drive', 4500),
(110, 'Webcam', 3000);

INSERT INTO Sale (bill_no, bill_data, cust_id, item_id, qty_sold) VALUES
(5001, '2026-01-10', 1, 101, 1),
```

(5001, '2026-01-10', 1, 105, 1),

(5002, '2026-01-12', 2, 102, 1),

(5003, '2026-01-15', 3, 103, 2),

(5004, '2026-01-18', 4, 108, 5),

(5005, '2026-01-20', 5, 104, 1),

(5006, '2026-01-22', 6, 110, 1),

(5007, '2026-01-25', 7, 107, 2),

(5008, '2026-02-01', 8, 109, 1),

(5009, '2026-02-05', 9, 106, 3),

(5010, '2026-02-10', 10, 102, 1);

Cust_id	cust_name
1	Amit Sharma
2	Priya Singh
3	John Doe
4	Sita Verma
5	Rajesh Kumar
6	Ananya Iyer
7	Vikram Rathore
8	Megha Gupta
9	Suresh Raina
10	Kavita Reddy
NULL	NULL

item_id	item_name	price
101	Laptop	55000
102	Smartphone	25000
103	Headphones	2500
104	Smartwatch	5000
105	Keyboard	1200
106	Gaming Mouse	1800
107	Monitor	12000
108	USB Drive 64GB	800
109	External Hard Drive	4500
110	Webcam	3000
NULL	NULL	NULL

bill_no	bill_data	cust_id	item_id	qty_sold
5001	2026-01-10	1	101	1
5001	2026-01-10	1	105	1
5002	2026-01-12	2	102	1
5003	2026-01-15	3	103	2
5004	2026-01-18	4	108	5
5005	2026-01-20	5	104	1
5006	2026-01-22	6	110	1
5007	2026-01-25	7	107	2
5008	2026-02-01	8	109	1
5009	2026-02-05	9	106	3
5010	2026-02-10	10	102	1
NULL	NULL	NULL	NULL	NULL

2. List all bills of the current date with customer names and item names.

```
SELECT
   S.bill_no,
   S.bill_data,
```

```
  C.cust_name,
  I.item_name,
  S.qty_sold
FROM
  Sale S
JOIN
  Customer C ON S.cust_id = C.Cust_id
JOIN
  Item I ON S.item_id = I.item_id
WHERE
  S.bill_data = CURDATE();
```

bill_no	bill_data	cust_name	item_name	qty_sold
5020	2026-03-01	Kavita Reddy	Smartphone	1

3. Display total bill amount using quantity and price.

```
SELECT
  S.bill_no,
  C.cust_name,
  I.item_name,
  S.qty_sold,
  I.price,
  (S.qty_sold * I.price) AS Total_Bill_Amount
FROM
  Sale S
JOIN
  Customer C ON S.cust_id = C.Cust_id
JOIN
  Item I ON S.item_id = I.item_id
```

bill_no	cust_name	item_name	qty_sold	price	Total_Bill_Amount
5001	Amit Sharma	Laptop	1	55000	55000
5001	Amit Sharma	Keyboard	1	1200	1200
5002	Priya Singh	Smartphone	1	25000	25000
5003	John Doe	Headphones	2	2500	5000
5004	Sita Verma	USB Drive 64GB	5	800	4000
5005	Rajesh Kumar	Smartwatch	1	5000	5000
5006	Ananya Iyer	Webcam	1	3000	3000
5007	Vikram Rathore	Monitor	2	12000	24000
5008	Megha Gupta	External Hard Drive	1	4500	4500
5009	Suresh Raina	Gaming Mouse	3	1800	5400
5010	Kavita Reddy	Smartphone	1	25000	25000
5020	Kavita Reddy	Smartphone	1	25000	25000

4. List customers who purchased products priced above 200.

```
SELECT DISTINCT
  C.cust_name
FROM
  Customer C
JOIN
  Sale S ON C.Cust_id = S.cust_id
JOIN
  Item I ON S.item_id = I.item_id
WHERE
  I.price > 200;
```

cust_name
Amit Sharma
Priya Singh
Kavita Reddy
John Doe
Rajesh Kumar
Suresh Raina
Vikram Rathore
Sita Verma
Megha Gupta
Ananya Iyer

5. Count the number of products purchased by each customer.

```
SELECT
  C.cust_name,
  COUNT(S.item_id) AS Number_of_Products
```

```
FROM
  Customer C
LEFT JOIN
  Sale S ON C.Cust_id = S.cust_id
GROUP BY
  C.Cust_id, C.cust_name;
```

cust_name	Number_of_Products
Amit Sharma	2
Priya Singh	1
John Doe	1
Sita Verma	1
Rajesh Kumar	1
Ananya Iyer	1
Vikram Rathore	1
Megha Gupta	1
Suresh Raina	1
Kavita Reddy	2

6. List products purchased by customer having Cust_id = 5.

```
SELECT
  I.item_name
FROM
  Item I
JOIN
  Sale S ON I.item_id = S.item_id
WHERE
  S.cust_id = 5;
```

item_name
Smartwatch

7. Create a VIEW showing bill_no, bill_date, cust_id, item_id, price, qty,sold, amount.

```
CREATE VIEW Sale_Detailed_View AS
SELECT
  S.bill_no,
  S.bill_data,
```

```
  S.cust_id,
  S.item_id,
  I.price,
  S.qty_sold,
  (S.qty_sold * I.price) AS amount
FROM
  Sale S
JOIN
  Item I ON S.item_id = I.item_id;
```

bill_no	bill_data	cust_id	item_id	price	qty_sold	amount
5001	2026-01-10	1	101	55000	1	55000
5002	2026-01-12	2	102	25000	1	25000
5010	2026-02-10	10	102	25000	1	25000
5020	2026-03-01	10	102	25000	1	25000
5003	2026-01-15	3	103	2500	2	5000
5005	2026-01-20	5	104	5000	1	5000
5001	2026-01-10	1	105	1200	1	1200
5009	2026-02-05	9	106	1800	3	5400
5007	2026-01-25	7	107	12000	2	24000
5004	2026-01-18	4	108	800	5	4000
5008	2026-02-01	8	109	4500	1	4500
5006	2026-01-22	6	110	3000	1	3000

8. Create a VIEW showing daily sales date-wise for the last one week.

```
CREATE VIEW Last_Week_Sales_View AS
SELECT
  bill_data,
  SUM(qty_sold * price) AS total_daily_sales
FROM
  Sale S
JOIN
  Item I ON S.item_id = I.item_id
WHERE
  bill_data >= DATE_SUB(CURDATE(), INTERVAL 7 DAY)
GROUP BY
```

bill_data;

	bill_data	total_daily_sales
▶	2026-03-01	25000